

Data Cleaning and Preprocessing Pipeline

Project Overview

Data Cleaning and Preprocessing Pipeline is a Python-based data engineering project developed to transform raw and unstructured datasets into clean, analysis-ready data.

Real-world datasets often contain duplicate records, missing values, incorrect data types, inconsistent formatting, and invalid entries. This project demonstrates how these issues can be identified and resolved using the Pandas library.

The final output is a cleaned dataset suitable for analytics, reporting, and machine learning applications.

Objectives

- Load and inspect raw datasets
 - Detect and remove duplicate records
 - Handle missing values using appropriate strategies
 - Correct incorrect data types
 - Standardize text and categorical values
 - Prepare analysis-ready datasets
 - Export cleaned datasets for further use
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Features

Data Ingestion

- CSV File Loading
- Dataset Inspection
- Basic Statistical Analysis

Deduplication

- Exact Duplicate Detection
- Duplicate Removal
- Redundancy Checking

Missing Value Handling

- Missing Value Identification
- Mean Imputation
- Median Imputation
- Mode Imputation
- Row Removal

Data Type Correction

- String to Numeric Conversion
- String to Datetime Conversion
- Category Standardization

Data Standardization

- Lowercase Conversion
- Whitespace Removal
- Category Mapping
- Format Consistency

Data Export

- Clean Dataset Generation
- CSV Output Creation

Technologies Used

Technology	Purpose
Python	Programming Language
Pandas	Data Cleaning
NumPy	Numerical Operations
Matplotlib	Visualization
Jupyter Notebook	Analysis
CSV Files	Dataset Storage

Installation

Clone Repository

```
git clone https://github.com/yourusername/Data-Cleaning-and-Preprocessing-Pipeline.git
```

Install Dependencies

```
pip install -r requirements.txt
```

Run Project

```
python src/data_cleaning.py
```

Workflow

Step 1: Load Dataset

Read raw customer data from CSV files.

Step 2: Remove Duplicates

Identify and remove repeated records.

Step 3: Handle Missing Values

Replace missing values using mean, median, or mode.

Step 4: Correct Data Types

Convert dates and numeric columns into proper formats.

Step 5: Standardize Formats

Normalize text values and category labels.

Step 6: Save Clean Data

Export cleaned dataset as:

cleaned_customers.csv

Sample Cleaning Operations

- Removed duplicate customer entries
 - Standardized customer names
 - Filled missing age values
 - Converted dates to datetime format
 - Corrected inconsistent region names
 - Removed unnecessary columns
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Testing Results

Test Case	Status
Dataset Loading	Passed
Duplicate Detection	Passed
Duplicate Removal	Passed
Missing Value Handling	Passed

Test Case	Status
Data Type Conversion	Passed
Text Standardization	Passed
CSV Export	Passed

Learning Outcomes

- Advanced Pandas Operations
- Data Quality Assessment
- Data Wrangling Techniques
- Missing Value Strategies
- Data Validation Methods
- Data Engineering Fundamentals

Future Enhancements

- Automated Data Quality Reports
- Interactive Dashboard
- Database Integration
- Large Dataset Processing
- Machine Learning Pipeline Integration

Screenshots

Raw Dataset

(Add Screenshot)

Data Cleaning Process

(Add Screenshot)

Final Clean Dataset

(Add Screenshot)

Author

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Internship Project – Data Cleaning and Preprocessing Pipeline

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